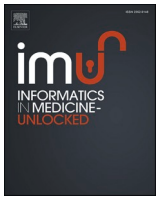
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Optimized real-time diagnosis of neglected tropical diseases by automatic recognition of skin lesions

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ABSTRACT

Background: Skin lesions associated with certain endemic neglected tropical diseases found in landlocked areas have irreversible effects on populations, contributing to population impoverishment. According to the World Health Organization (WHO), there were 127558 new leprosy cases detected in 2020, with about 8 629 children under the age of 15 affected with approximately 7 198 suffering from grade 2 disabilities (G2D). In the same year, about 1 million new leishmaniasis cases were reported globally, whereas the number of suspected Buruli ulcer cases reported globally in one year was around 5000, with the majority of patients under the age of 15. Current diagnostic methods are unfortunately expensive, not suitable for endemic areas, and sometimes not very effective. According to the SDG Target 3.3, WHO plans to end the neglect tropical skin diseases (NTDs) by 2030. According to them, this required an early diagnosis. Our work belongs to this objective.

Methods: This paper presents an optimized diagnostic approach for neglected tropical skin diseases (NTDs) by automatically identifying skin lesions in their early phase. We chose three main skin diseases with similar early lesion development. First of all, we have built a dataset of images of these lesions. After accurately detecting the location of the skin lesion, we delineate the edges of the lesion. In addition, we apply a series of filters to detect and extract relevant features from these lesions. Finally, the detected features are fed into a support vector machine (SVM) optimized by a black hole algorithm (BHO) to be classified in real time (less processing time).

Results: Our model achieves a global classification accuracy of 96%, 94% specificity, an F-SCORE of 89%, a recall of 90%, and a sensitivity of 92% of the images under different conditions and a processing time that is lower than other classical algorithms. These results are superior to current diagnostic methods such as direct smear examination with a low sensitivity (<60%), in vitro culture with a low sensitivity (20–60%), and PCR for IS2404 with high sensitivity and specificity (>90%) (but which requires sophisticated laboratories and competent personnel). These results can improve the sanitary situation of the population and enable the development of efficient Computer-Aided Diagnosis (CAD).

Conclusion: The results obtained in this research produce the best solution compared to other algorithms and can be used to make a real time diagnosis of NTDs.

Trial registration: "Not applicable"

1. Introduction

Neglected tropical skin diseases have numerous consequences for over a thousand million people in tropical and semi-tropical countries, costing developing countries too much money each year [1,2]. The majority of these diseases have a cutaneous manifestation. The diagnosis of NTDs, if not made early, often leads to irreversible consequences. However, these diseases, when detected early, are easy to treat. Unfortunately, NTDs generally occur in remote areas that lack basic health infrastructure or qualified personnel to diagnose them in their early stages.

Buruli ulcer is an infectious skin disease caused by Mycobacterium Ulcerans. Although the mode of transmission is unknown, there is often a link with the presence of rivers, streams, or swampy areas. Initial skin abnormalities often appear as raised nodules that later ulcerate. The ulcer is usually single, large, and has a yellowish surface with a moist,

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Table 1
Summary of diagnostic methods.

Method	Advantages	Disadvantages
Direct smear examination	- Can be performed on swab, FNA or biopsy samples. - Easy to perform locally - Does not require expensive materials and equipment - Rapid results	- Low sensitivity (<60%) - Requires trained personnel - Requires strict quality control
In vitro culture of Mycobacterium Ulcerans	- Can be performed on swab, FNA or biopsy samples. - Only method capable of distinguishing viable from non-viable organisms. - Can be used to monitor patient response to mycobacterial therapy.	- Requires a sophisticated laboratory - Requires trained staff - Slow results (at least 6–12 weeks) - Low sensitivity (20–60%) - Not useful for immediate patient care - Requires strict quality control
PCR for IS2404	- Can be performed on swab, FNA or biopsy samples (fresh or paraffin embedded). - Rapid results - High sensitivity and specificity (>90%)	- Requires a sophisticated laboratory -Costly to carry out - Requires highly trained staff - Requires strict quality control - Cannot distinguish between viable and non-viable organisms.
Histopathology	- Fairly fast results - High sensitivity (about 90%) - Useful for differential diagnosis and monitoring response to treatment	- Requires a sophisticated laboratory -Costly to carry out - Requires highly trained personnel - Requires an invasive procedure (i.e. biopsy)

reddish base. Another disease that begins with nodules or plaques in its early stages that later also ulcerate is cutaneous leishmaniasis (CL). CL is an infection of the skin caused by a protozoan of the genus Leishmania. The infection is spread by biting insects, sandflies belonging to the genera Phlebotomus, Lutzomyia, or Psychodopygus. Reservoir hosts can be wild or domestic animals as well as humans. Like leishmaniasis and Buruli ulcer, leprosy usually begins with dry plaques or nodules that may be slightly paler in color than the rest of the skin. Leprosy is a systemic infectious disease caused by the bacterium Mycobacterium leprae. It mainly affects the skin and nerves. Although much of the disability caused by leprosy is due to the destruction of the nerves, the skin is also affected in the early stages of infection. The disease is transmitted directly through close and prolonged contact.

Therefore, it is important to equip health services with a tool that can quickly identify a neglected skin-related disease and automatically refer it to specialized centers. The diagnosis of these diseases often requires a very sophisticated technical platform that is not accessible in these endemic areas, and the various methods used with their advantages and disadvantages can be summarized in Table 1.

According to the World Health Organization (WHO), NTDs can be effectively controlled by administering antibiotics if they are detected in their earliest stages. But the lifestyle of the population (usually coming to the hospital when the disease is already in its most advanced stage and thus requiring costly intervention such as surgery), coupled with the precarious situation of the endemic infected areas where they live, especially in Cameroon, which lacks hospitals, qualified health personnel and electrical infrastructure, makes early diagnosis of these diseases difficult. Therefore, it is important to equip frontline staff with a tool that can quickly identify a neglected skin-related disease just by taking a picture with a computer-aided diagnosis (CAD).

There have been several successful projects, both in academia and industry, on the recognition of skin diseases such as skin cancers [3,4], but also some work on skin-related NTDs. This section focuses on existing work on automatic skin lesion recognition technology for neglected tropical diseases using machine learning approaches.

In their article [5], Rui Hu et al. work on a replacement segmentation theme in order to observe the border of Buruli lesion lesions in cross-polarized dermoscopic images obtained beneath white lightweight illumination. The authors in this research segment the images of the disease in the ulcer phase, which does not solve the problem of early detection of the disease.

In 2018, Marc Górriz et al. presented a procedure for automating the detection process based on a deep learning approach. The authors built and trained a neural net model that successfully segments leishmaniasis parasites and classifies them [6]. In 2022, Mojtaba Zare et al. developed a synthetic intelligence-based formula for the automated diagnosis of kala-azar [7]. The authors used the Viola-Jones formula to develop a kala-azar parasite detection system. The work of [6,7] is based on a dataset of microscopic images of the germs and not of the skin lesions themselves. This does not solve the real problem of the affected areas: the lack of equipment to identify the germs.

Márcio Luís Moreira De Souza et al. developed an application for skin Hansen's disease detection supported by machine learning (AI) for the diagnosis of Hansen's disease, particularly for communities distant from major diagnostic centers. The metrics' sensitivity and specificity reached 93.97% and 8.09%, respectively [8].

Raquel R. Barbieri launched research into an AI-driven skin image-based leprosy diagnostic assistant [9]. The authors used Here, an AI-based leprosy diagnostic assistant based on a combination of skin images and clinical data collected through a standardized process. A high classification accuracy (90%) and an area under the curve (AUC) of 96.46% for the diagnosis of leprosy were obtained. The results obtained in the literature with the SVM seem to be all below 95%. The question arises: is it not possible to increase this performance?

According to some research, metaheuristics algorithms optimize SVM by improving their time and classification performance. As a result, nature-inspired optimization tools provide a way to automate the process of finding the best solution. for example, the authors of [13] propose a new Nature Inspired Optimization Tools for SVM optimization, which provides an approach to automate the process of finding the best possible solution for parameter selection, allowing the user to quickly obtain several sets of good solutions.

But in practice, in this last study, each metaheuristic generates several non-dominant sets that are used to obtain a general set of non-dominant solutions. This introduces a problem of parallelism in data processing. For that, the authors of [14] compare the strength of metaheuristic algorithms on SVM to evaluate the stability, convergence rate, feature selection ability, and classification accuracy. Particle swarm optimization (PSO), Genetic Algorithms (GA), and artificial bee colony algorithms (ABC) are used to optimize the SVM. GA-SVM obtained the highest overall average accuracy (91.77%), followed by ABC-SVM (88.73%) and PSO-SVM (86.65%). It is highlighted that, while the accuracy is quite high, it may yet be improved. The authors of [15] demonstrated this by optimizing an SVM with a differential flower pollination algorithm to classify corroded and intact pipe surfaces across a set of 2000 image samples. The results reveal that the proposed method is well suited to the task at hand, with an accuracy rate of 92.81%. The authors of [16] have confirmed that result by applying Grey Wolf Optimization (GWO-SVM), Whale Optimization Algorithm (WOA-SVM), and Moth Flame Optimization (MFO-SVM) to the SVM to predict the annual performance of the TBM. The accuracy of 96.24% confirms that the hybridization of the SVM with metaheuristics is very promising in increasing their performance.

In recent years, with the rapid development of artificial intelligence, computer-aided diagnosis (CAD) has also seen a series of advances, particularly in the field of medical imaging. Examples include the



Fig. 1. Evolution of Buruli ulcer.



Fig. 2. Evolution of leishmaniasis.



Fig. 3. Evolution of leprosy.

modelling of current public health problems such as COVID-19 [19] and mental health education activity in universities [25]. As these disease models are generally non-linear, artificial intelligence models have often been developed, such as the authors of [20], who succeeded in modeling and solving a non-linear system of influenza disease (NIDS) using Morlet wavelet neural networks (MWNN) optimized by hybrid global/local search approaches. The analysis of approximate analytical solutions is also used after modeling, as done by the authors of [21,22] for solving vector disease models and Covid19. The difficulty of implementing these non-linear mathematical models coupled with the complexity of health information systems has made traditional methods obsolete, and simple mathematical-statistical methods are almost obsolete due to problems caused by the explosive growth of data and information. For this reason, data science methods have emerged [23–26].

The questions that this work would like to answer are: Can the optimization of an SVM by a black hole algorithm increase its accuracy and speed?

The main objective considered in this paper is to propose an optimized SVM-BHO classifier. This classifier can be used on a mobile phone to detect skin lesions automatically and can help healthcare workers quickly identify lesions in their early phase in real time.

Figs. 1–3 show a sample of three skin diseases from their early to more severe stage. The images to be identified are those characterized by the early phase of the diseases, as they are easy to treat, usually with antibiotics. In this study, we will limit ourselves to these three diseases, namely Buruli ulcer, leishmaniasis, and leprosy.

2. Methodology

This section presents the methodology set up for the system, which is divided into two main modules: the extraction of the parameters to build our dataset, which here are the fractal dimension and the texture of the

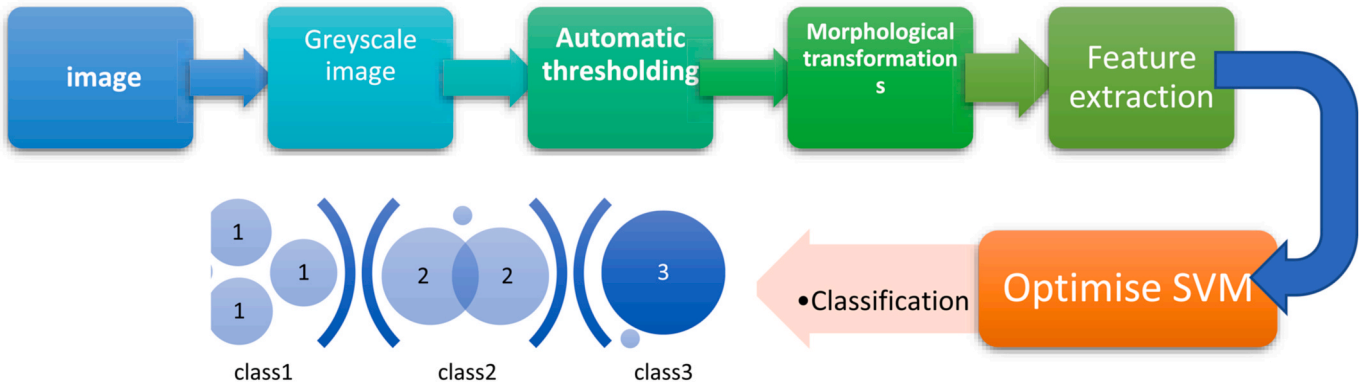


Fig. 4. Overview of image processing and classification.

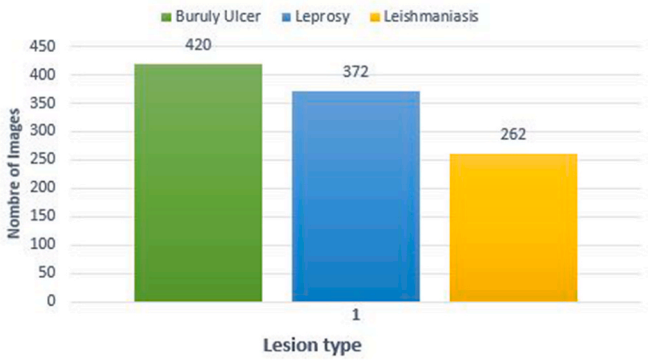


Fig. 5. Distribution of the dataset.

grey scale image; and the recognition modules performed by an optimized SVM in a web application. In Fig. 4, we present the detection algorithm and in the rest of this section we discuss these modules separately.

2.1. Dataset

In order to diagnose neglected tropical diseases in their early phase, we collected a set of skin lesions of the three main diseases in Cameroon: leprosy, leishmaniasis, and Buruli ulcer [10–12]. Fig. 5 shows the number of images per disease. We counted 372 images of lesions due to leprosy, 420 dues to BU and 262 dues to cutaneous leishmaniasis. This makes a total of 1054 images to be processed. The study was approved by the Cameroon National Ethics Committee under Authorization No. 041/CNE/DNM/09. All data were anonymized, and the handling and analysis of the data were strictly confidential [18]. Medical photography has been meticulously followed, as it is a specialized genre that requires meticulous control of camera and environmental settings to achieve optimal quality images that accurately depict the human body. The fundamentals of photography have been considered, including exposure, ISO, aperture, shutter speed, focus, focal length, and composition.

2.2. Pre-processing techniques

This phase consists of pre-processing the images in order to standardize them to bring out the best characteristics of each lesion. This phase takes place in three stages as shown in Fig. 6:

a Transformation to Grey level

To cancel the effects of gamma compression before calculating the greyscale luminance, it is necessary to apply the opposite operation, gamma expansion:

$$C_{lineaire} = \begin{cases} \dfrac{C_{srgb}}{12.92}, & Csrb \leq 0.04045 \\ \left(\dfrac{C_{srgb} + 0.055}{1.055}\right)^{2.4}, & C_{srgb} > 0.04045 \end{cases} \quad (1)$$

where C_{srgb} the three base colors primaries R_{srgb}, G_{srgb} and B_{srgb} saturation, in the range of [0,1] and C_{linear} is the linear intensity value R_{linear}, G_{linear}, and B_{linear}, belonging to [0,1]. Then we can calculate C_{linear} as given by Ref. [17].

b. **Automatic thresholding**

In Otsu Thresholding, a threshold value is not chosen but is determined automatically. This results in a binary image.

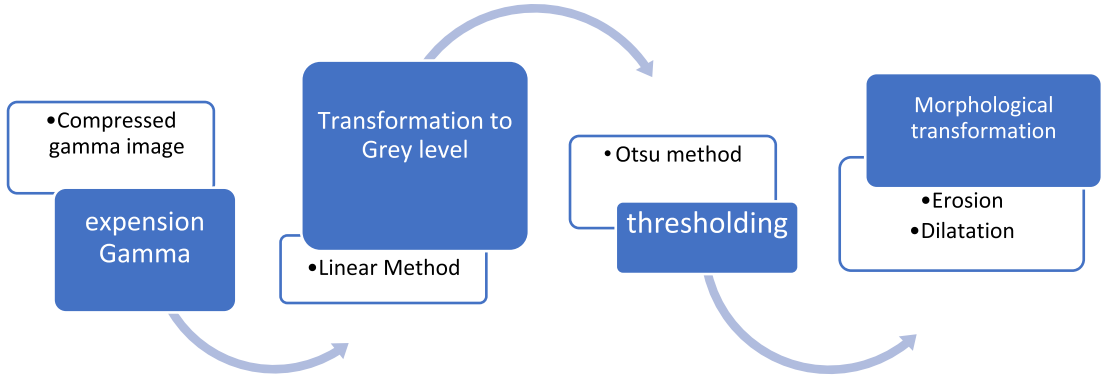


Fig. 6. Preparation of images.

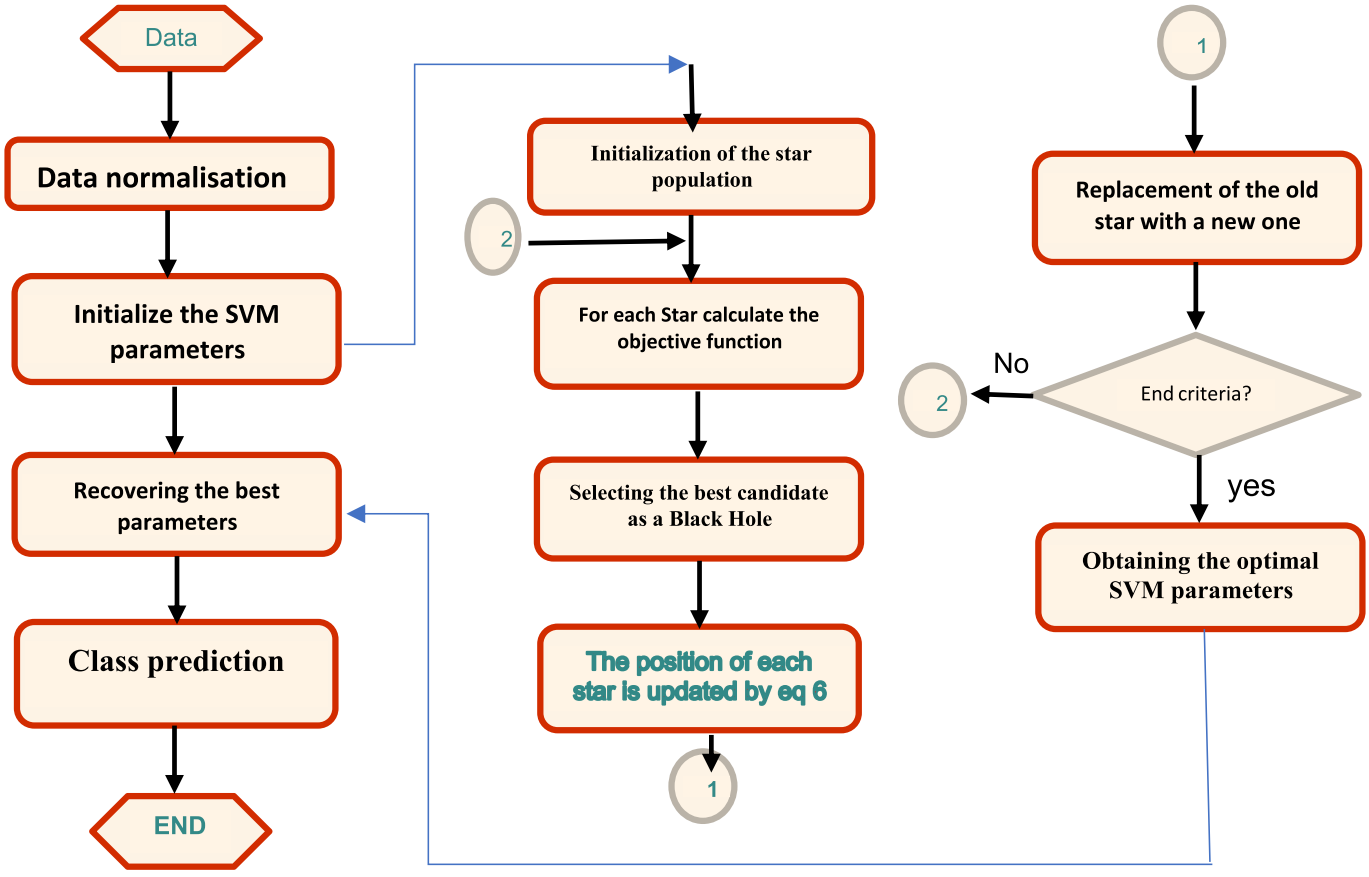


Fig. 7. The black hole SVM optimization algorithm.

c. Morphological transformation

Morphology is a technique that allows a matrix (and therefore an image) to be transformed by moving a structure over each pixel of the image. When a "white" pixel is visited, the structure can perform an operation: dilation and erosion.

2.3. Classification

The main objective of SVM is to find the optimal hyperplane that linearly separates the data points into two components by maximizing the margin. Mathematically, this involves classifying the data around a generally straight line and minimizing the distance of all classes from this dividing line.

The associated objective function is as follows:

$$min_w\ \lambda w^2 + \sum_{i=1}^{n}(1 - y_i x_i,\ w). \tag{2}$$

The minimization of this function will be done by a metaheuristic which is the black hole algorithm (BHO). With equation (3) describing the hyperparameters of the SVM.

$$w = w + \eta(y_i x_i - 2\lambda w). \tag{3}$$

The above optimization operation can be solved with the SMO (sequential minimization optimization) or Gradient Descent algorithms. In our case we apply a black hole algorithm (BHO), which we compare with the SMO and gradient descent contained in the python library scipy.

2.4. Black hole algorithm optimization

A region contour is observed around the black hole called the horizon of even which radius is commonly called the Schwarzschild radius and can be processed mathematically using equation 4

$$R = \frac{2GM}{C^2}. \tag{4}$$

G is the gravitational constant and is equal to 6.67408×10^{-11}, M is the mass of the black hole and C is the speed of light and is equal to 3×10^{8} m/s.

$$R = \frac{f_{bh}}{\sum_{k=1}^{n} f_k}. \tag{5}$$

f_{bh} is the cost or fitness value of the best star; n is the total number of stars in the population, f_k is the cost of the star (k). If the distance between the star and the black hole is less than or equal to R, it will be deleted and a new star will be arbitrarily generated and added to the population. The movements of the stars are obtained by updating their position to a new position closer to the black hole by applying the equation below.

$$X_i(t+1) = X_i(t) + r^*(X_{BH}(t) - X_i(t)). \tag{6}$$

In this case, X(t) represents the location vector of star (i) at iteration t, XB(t) represents the location vector of the black hole at iteration t, and r is a number between [0,1]. At each iteration, all stars must update their location in the search space, affected by the gravity power of the current best solution. The optimal search for the solution of this dual problem will be as shown in the Fig. 7 below.

Initialize the population with randomly generated stars
for $i = 1$ to the total number of stars
    calculate the ith star's fitness value and save in an array (f)
end
Determine the black hole which is the star with the highest fitness value
While (Criteria for max iteration T or convergence are not met) do
    for $i = 1$ to number of stars
        $x_i^{new} = x_i^{old} + rand * (x_{BH} - x_i^{old})$ /*The position of the stars change */
        Assess the new star's fitness value (x_i^{new})
        if the new star's fitness value (x_i^{new}) < fitness value of x_{BH} **then**
            $x_{BH}=x_i^{new}$ /* when a star enters a lower-cost position than the black hole (BH), the location of the BH is changed with the star*/
        end
        Update the star's fitness value (f_i) with the new value
        Update the fitness value in the array (f)
        Calculate the radius of the event horizon $R = \frac{f_{BH}}{\sum_{i=1}^{nstars} f_i}$
        if $\sqrt{(x_{BH} - x_i^{new})^2} < R$ **Then**
            Reinitialize the star randomly in the search space
        end
    end

Fig. 8. The black hole SVM optimization algorithm.

3. Proposed method's theoretical analysis

In this section we will present a numerical example of SVM optimization to explain how the algorithm works. Suppose we have a dataset $\{x_i\, y_i\}_{i=1\ldots n}$ where x_i *et* y_i represent the inputs and outputs respectively. The ultimate goal of SVM is to draw the linear decision function given by equation (7):

$$d(x) = w, \Phi_i(x) + b \tag{7}$$

The hyperparameters w and b represent respectively the weight and the constant to be estimated by the dataset. Φ is a non-linear function that maps the input features to a higher feature space. The symbol $\langle \ldots \rangle$ indicates the scalar product.

The problem is to minimize the following function:

$$F(C) = \frac{C}{n}\sum_{i=1}^{n} L_\varepsilon(d(x_i), y_i) + \frac{1}{2}w^2 \tag{8}$$

$L_\varepsilon(d(x_i), y_i)$ is taken as the loss function shown in equation (9) below

$$L_\varepsilon(d(x), y) = \begin{cases} |d(x) - y| - \varepsilon, & |d(x) - y| \geq \varepsilon \\ 0, & sinon \end{cases} \tag{9}$$

The function to be minimized can be simplified by introducing two soft parameters η et η* to equation (8) which becomes our new objective function (Eq.(10)) with constraints according to equation (11).

$$F(w, \eta^*) = C\sum_{i=1}^{n}\left((\eta_i + \eta^*_i) + \frac{1}{2}w^2\right. \tag{10}$$

$$\begin{cases} y_i - w, x_i - b \leq \varepsilon + \eta_i \\ w, x_i + b - y_i \leq \varepsilon + \eta_i^* \\ \eta_i, \eta_i^* \geq 0 \end{cases} \tag{11}$$

The update of the weights will be done by using equation (6) as follows.

$$w_i(t+1) = w_i(t) + r^*(w_{BH}(t) - w_i(t)). \tag{12}$$

$$\eta_i(t+1) = \eta_i(t) + r^*(\eta_{BH}(t) - \eta_i(t)). \tag{13}$$

Knowing that the update condition is given by the condition

$$R = \frac{F_{bh}}{\sum_{k=1}^{n} F_k}.$$

C represents a penalty for a prediction error that is greater than ε, While η_i *et* η_i^* re variables that measure the cost of error during learning.

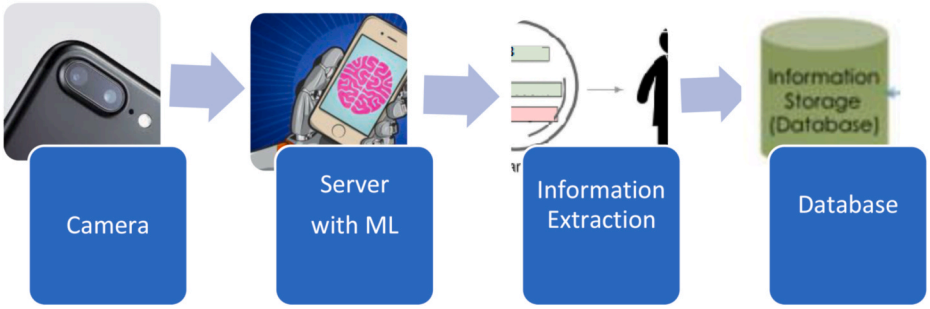


Fig. 9. Architecture of the NTDs management system.

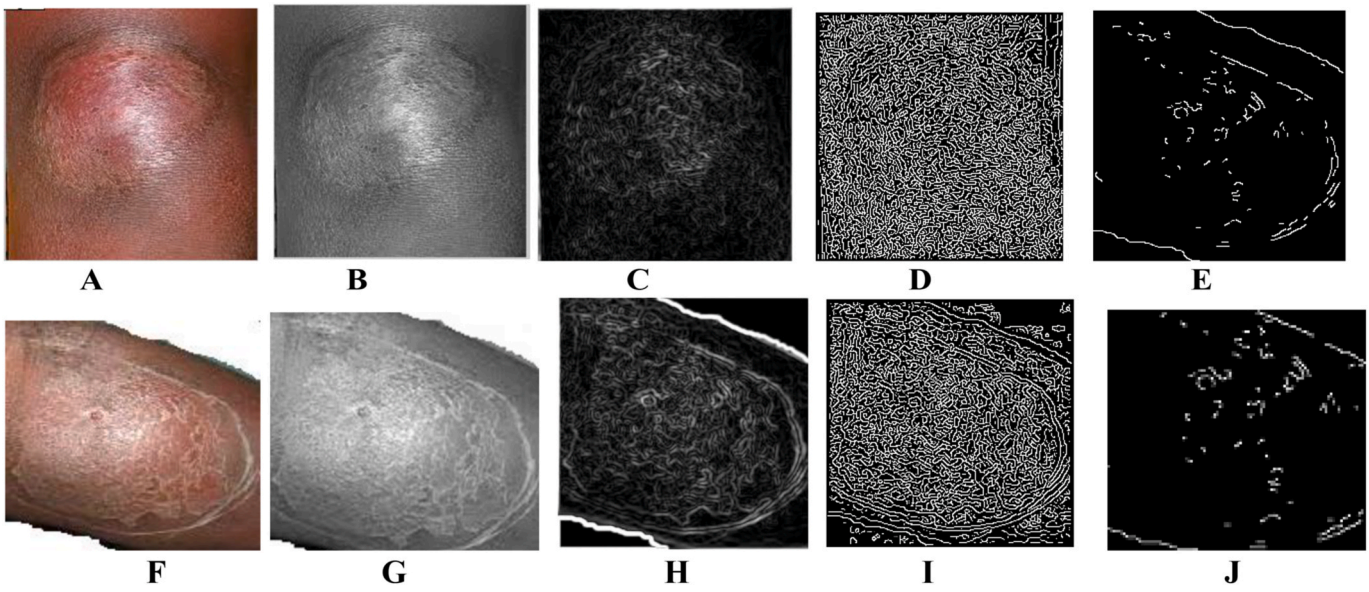


Fig. 10. Pre-processing of the BU lesion.

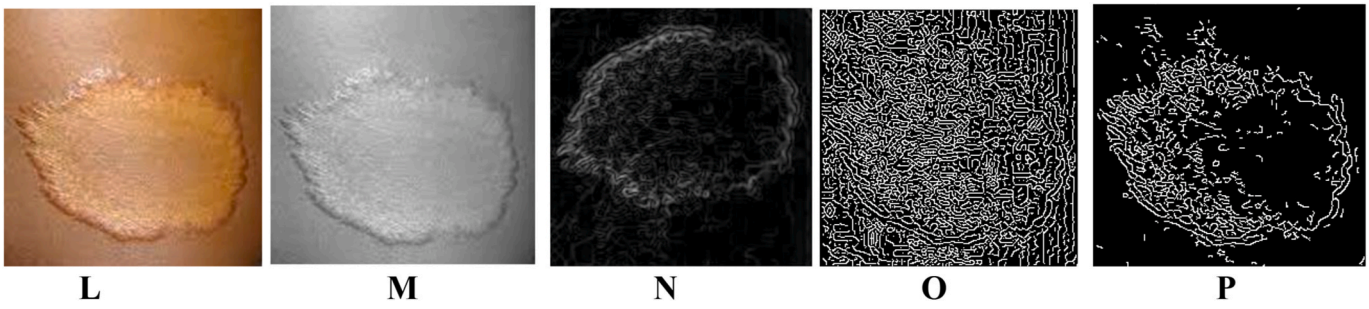


Fig. 11. Pre-processing of the leprosy lesion.

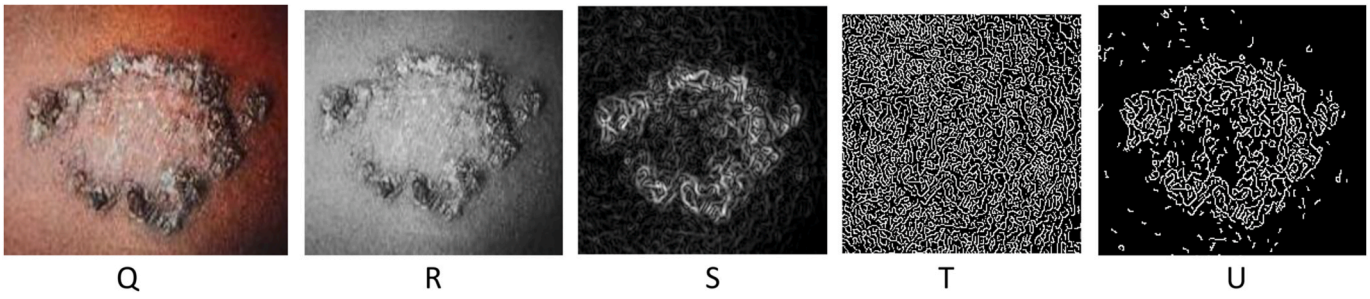


Fig. 12. Pre-processing of the LEISHMANIOSIS lesion.

The determination of the parameters w and b will be done by minimizing the objective function under its constraints by our metaheuristic following the pseudocode represented by Fig. 8 below (see Fig. 8).

4. Deployment and integration

This section presents the details of the implementation of our web-based diagnostic model. The lesion detection and classification modules are implemented using the Python programming language. As illustrated in the figure below, the mobile application allows users to take a picture of the lesion on the skin. Once the image is captured, the application uploads it to our automatic recognition model as illustrated in Fig. 9.

5. Results

5.1. Image pre-processing and processing

The results of the grey level transformation, automatic thresholding using the OTSU method, and mathematical morphology are presented here. We have the lesions of BU (A and F), leprosy (L), and leishmaniasis (Q). B and G are the grey level transforms of BU while M and R are those of leprosy and leishmaniasis respectively. C and H represent the image obtained after automatic thresholding of B and G, respectively. Similarly, N and S are obtained after thresholding M and R. The dilation operation allowed us to obtain D, I, O, and T, and finally, the erosion gives us the final image characterizing each lesion as illustrated in

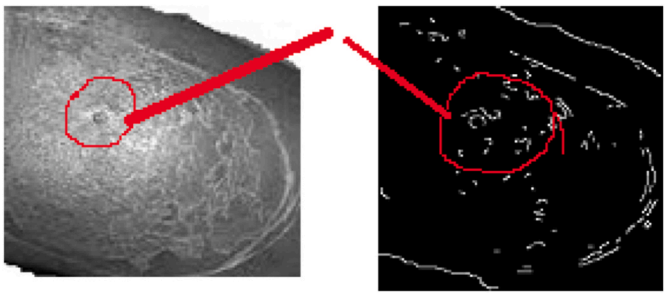


Fig. 13. Visual characterization of the BU lesion after pretreatment.

Figs. 10–12.

At first glance, we notice a possibility of visually distinguishing the various lesions. For the BU lesions after erosion, we see that the white pixels seem to converge towards a point which we believe is the likely starting point of the ulcer (wound). This is shown in the Fig. 13 below in red.

5.2. *Extraction of characteristics for classification*

The extraction of the parameters was done by applying a series of filters (Canny, Sober, Gabor, and Robert) on the preprocessed images to extract, among others, the average texture, the standard deviation of the texture, the average fractal dimension and its standard deviation, the average of the homogeneous zones, and the edges. The data resulting from this processing can be downloaded through the link https://github.

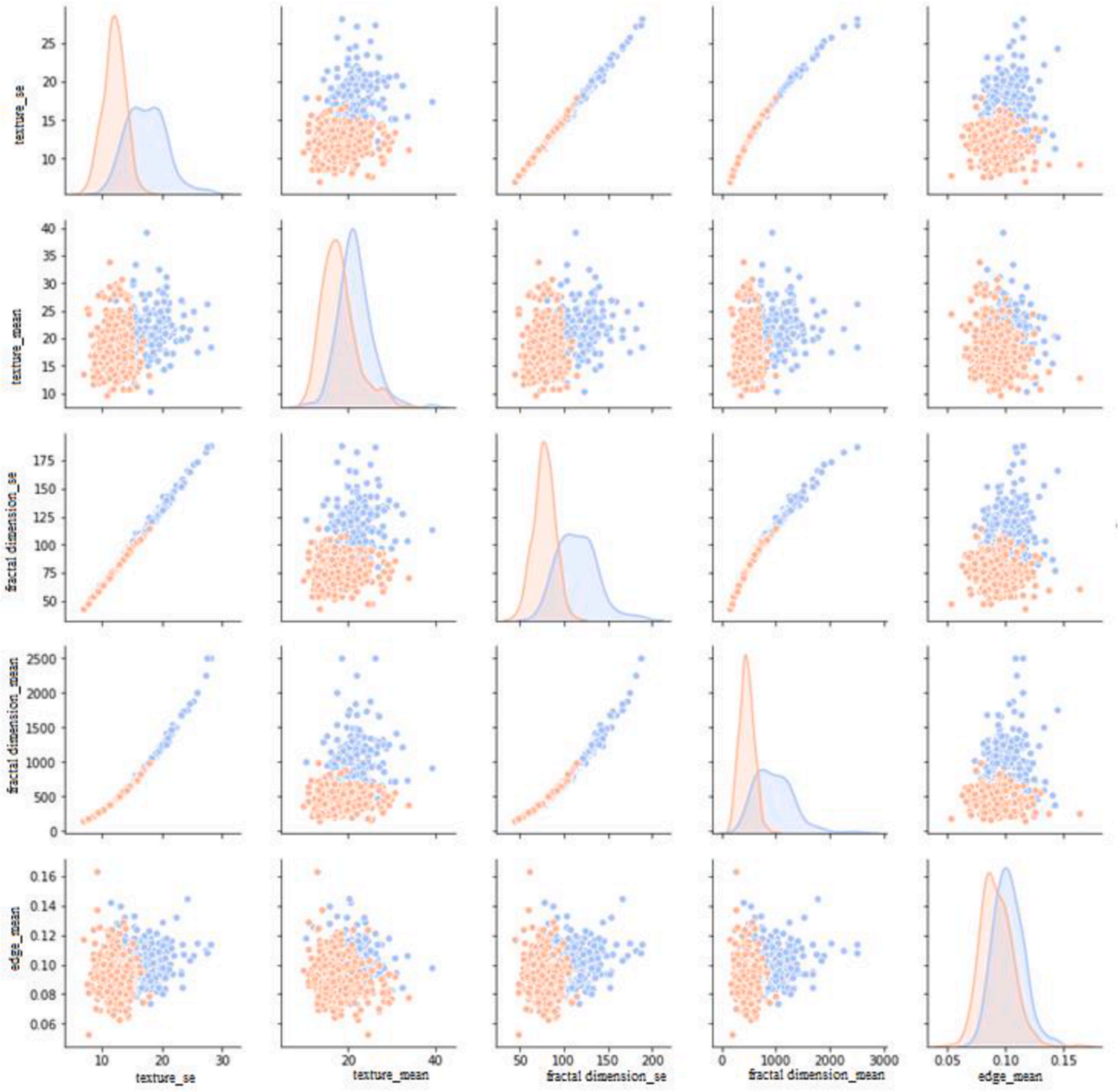


Fig. 14. Two by two view of the extracted features.

Table 2
Classification metrics comparison.

Metrics %	KNN	ANN	SVM	BHO-SVM
Accuracy	84	94	94	96
Specificity	67	89	94	94
F-Score	66	85	86	89
Recall	97	84	87	90
Precision/Sensitivity	88	77	97	92

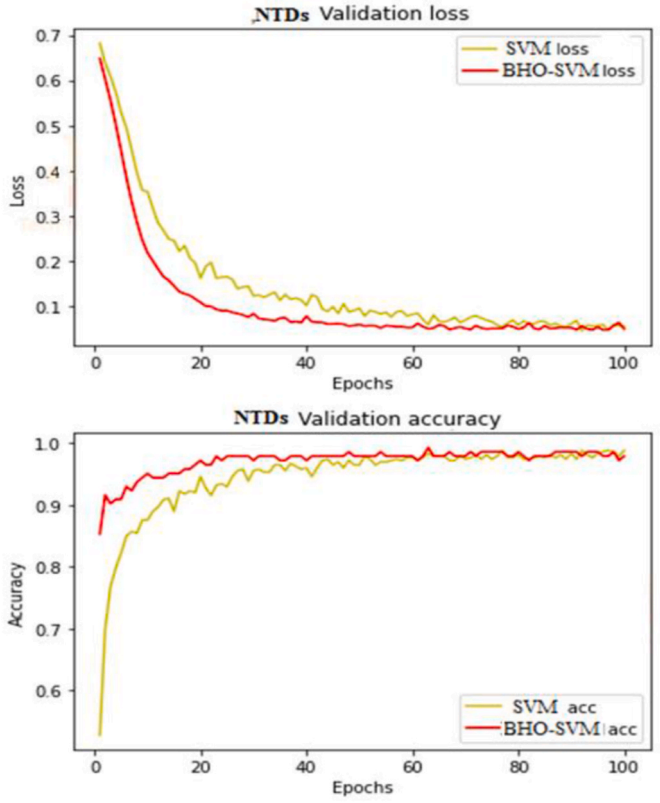


Fig. 15. Performance graphs.

com/dodo12/MTNskin-Diagnosis. The code to extract these parameters is also available here. The Fig. 14 below shows a two-by-two view of the extracted features.

The two-color scatter plot shows Buruli ulcer positive cases in red and negative cases in blue. We can clearly see a boundary between these data sets. This promises a good classification rate.

5.3. Classification metrics

We found that our model achieves a classification accuracy of 96% for the recognition of the skin lesions studied. We also calculated the overall classification accuracy by other simple algorithms like KNN, decision trees, and SVM. We found that our model achieves a global classification accuracy of between 92 and 96%. This is illustrated in the Table 2 below.

This table confirms what we said earlier: our dataset is subject to a better classification rate. We observe that our BHO-SVM algorithm performs better in the majority of the chosen metrics. The accuracy performance curves after 100 iterations as illustrated by Fig. 15.

5.4. Processing time

Since our automatic NTDs skin lesion recognition application must operate in real time, the processing time for detection is a critical factor

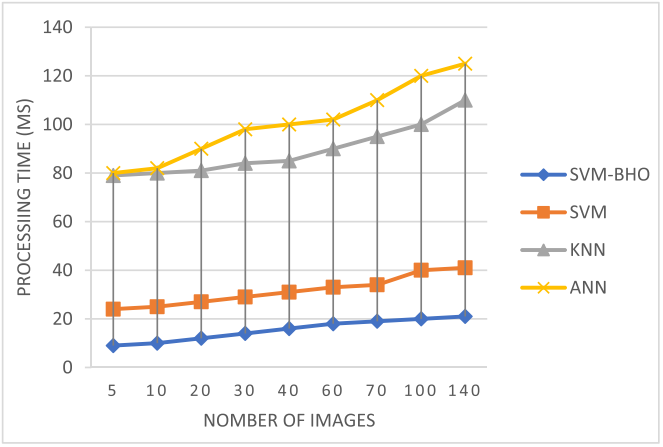


Fig. 16. CPU processing time of models 6.

Table 3
Summary of reference works on NTDs.

Reference	Method	Disadvantage
Rui Hu et al. [5]	SVM (FC)	- Very small number of images - Diagnosis in the ulcer phase, i.e. late.
Marc Górriz et al. [7]	Viola-Jones algorithm	Very low performance
Márcio Luís Moreira [8]	random forest algorithm	Study data are only clinical and environmental and therefore difficult to collect in patients who are generally illiterate
Raquel R Barbier [9]	CNN-based AI algorithm	Only works on one disease.
***Nyatte et al.**	BHO-SVM	Only image dataset available. we need to add metadata such as environmental characteristics and clinical data

in evaluating different image recognition models. We found that our model, on average, needed only about 27 ms to detect lesions. We used an HP EliteBook 835 G8 computer. Windows 10 Professional, i7 processor, 16 GB RAM, and 256 GB SSD. The processed result is shown by Fig. 16 below.

6. Discussion

In the literature, some authors have worked on the classification of neglected tropical diseases with a view to developing CAD (Computer-Aided Diagnostic) tools for neglected tropical diseases. Table 3 below provides a summary of these works with the basis, the methods used, and the drawbacks, in our opinion, of these works.

A comparison of the performance values shows that our model performs better and also introduces another metric, which is the treatment time. In Table 4 below, which presents the various diseases considered by the authors in their study, we are the only ones to have studied all three diseases at the same time for the purpose of diagnosis.

To the question of what proportion of predicted positives are really positives? The answer of 96% is excellent and is the best in the literature for similar problems. To the question, what proportion of true positives are correctly classified? The answer once again is the best, i.e., 90%.

The Fig. 17 shows that we are the only ones to evaluate the results over five performances. Our results are the best. The sensitivity and specificity of [8] are very close to our results despite the fact that the author has a very large image data set associated with the metadata. We believe that by adding clinical and environmental data to our dataset we can increase the performance.

Table 4
Comparison of results with the literature.

Author	Lesion type			Dataset		Performance				
	Buruli ulcer	Leishmaniasis	Leprosy	Images	Metadata	Accuracy	Specificity	F-score	recall	sensitivity
Rui Hu et al. [5]	yes	no	no	26	NO	85. 70%				
Márcio Luís Moreira [8]	no	yes	no	300	NO	50%	87.09%,		65%	93.97%
Raquel R Barbier [9]	no	no	yes	1229	587	90%				
***Nyatte et al.**	**yes**	**yes**	**yes**	**1054**	**NO**	**96%**	**94%**	**89%**	**90%**	**92%**

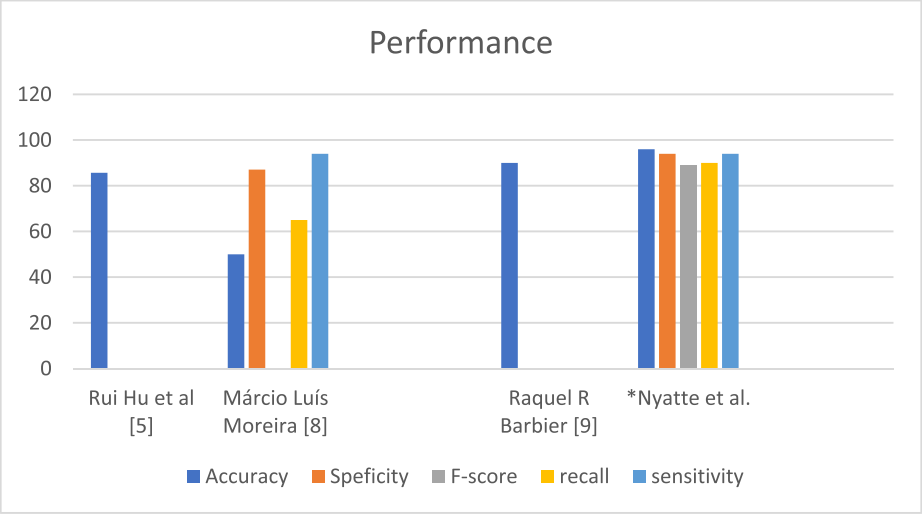


Fig. 17. Evaluation of the results obtained compared to the literature.

7. Conclusions

This paper presents the design and implementation of a management system for neglected tropical skin diseases. We believe that the proposed system will enable healthcare workers to automate and make reliable diagnoses during the early stages of diseases. The proposed approach for the automatic lesion recognition model consists of three phases: detection, segmentation, and classification. We tested our system with a dataset of different shapes and sizes, taking into account cluttered backgrounds, low contrast images, and various lighting conditions. Our model achieves a global classification accuracy of 96%, 94% specificity an F-SCORE of 89%, a recall of 90% and a sensitivity of 92 of the images under different conditions and a processing time that is lower than other classical algorithms. These results are superior to current diagnostic methods such as Direct smear examination with a low sensitivity (<60%), In vitro culture with Low sensitivity (20–60%), PCR for IS2404 with High sensitivity and specificity (>90%) with sophisticated laboratory. These results can improve sanitary situation of the population as is very easy to easy to handle and less expensive. For future work, we will augment our image dataset using data augmentation techniques, but we will also combine other types of data, mainly clinical and environmental, to assess the results of such changes.

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Availability of data and materials

Data and resources are available. On reasonable request, the datasets used in the current investigation are accessible from the corresponding author.

Authors' contributions

All authors participated in formulating the search strategy and identifying relevant studies. [1]**NS** wrote the first draft of manuscript and all authors critically reviewed and this last version was approved.

Ethics approval

Not applicable.

Publication consent

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

SDG Global Sustainable Development Goal
SVM Support Vector Machine;
BHO Black hole optimization
AI Artificial Intelligence
BU Buruli Ulcer
TNM Tumor Node Metastasis
WHO World Health Organization

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